Supplementary material

Machine learning prediction of pushover-based TBEC 2018 performance states of gravity-sized steel moment-resisting frames

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About This File

This file supports the article named above. Sections, equations, figures and tables without the prefix S belong to the article, and the sections follow the order of the article.

S1. Distributions of the Input Variables

Figure S1 shows how the 21 inputs of Table 2 are distributed in the database. Except where the height limit intervenes, the distributions reflect the sampling design of Section 2.1. Bay width, both story heights and both line loads are sampled continuously and appear nearly uniform. Story and bay counts, steel grade and soil class are discrete and keep their stratified shares, except that the 28 m height limit eliminated most of the tallest candidates. As a consequence, eight-story frames make up 6.7% of the population, while each of the other story counts makes up 15.3% to 15.6%, and large typical story heights are less frequent. Total height, total width and seismic mass are computed from several sampled variables, which skews their distributions.

The section dimensions need one remark. Columns and beams are standard European hot-rolled profiles, yet the models never receive a profile name. They receive the eight dimensions of Table 2, and because a rolled-section catalog offers a finite set of shapes, these dimensions form discrete clusters in Figure S1. The bars give the number of frames per bin or category, and for the hazard level, each frame is counted once at each of the two levels.

S2. Design Spectra of the Site

Every analysis takes its seismic demand from the horizontal elastic design spectrum of TBEC 2018 at the site, evaluated for its soil class and hazard level, and Figure S2 shows these spectra for the five soil classes. The long-period branch rises as the soil becomes softer, whereas the plateau is highest for ZC, and both rise as the hazard level changes from DD-2 to DD-1. Through these spectral changes, soil class and hazard level influence the target displacement and, through it, the label. The two inputs also enter the gravity state, because S_{DS} scales its vertical earthquake term (Section 2.2). The ZA spectra are dashed in Figure S2,

as they coincide with the ZB spectra at longer periods and, at DD-1, with the ZE spectrum on the plateau.

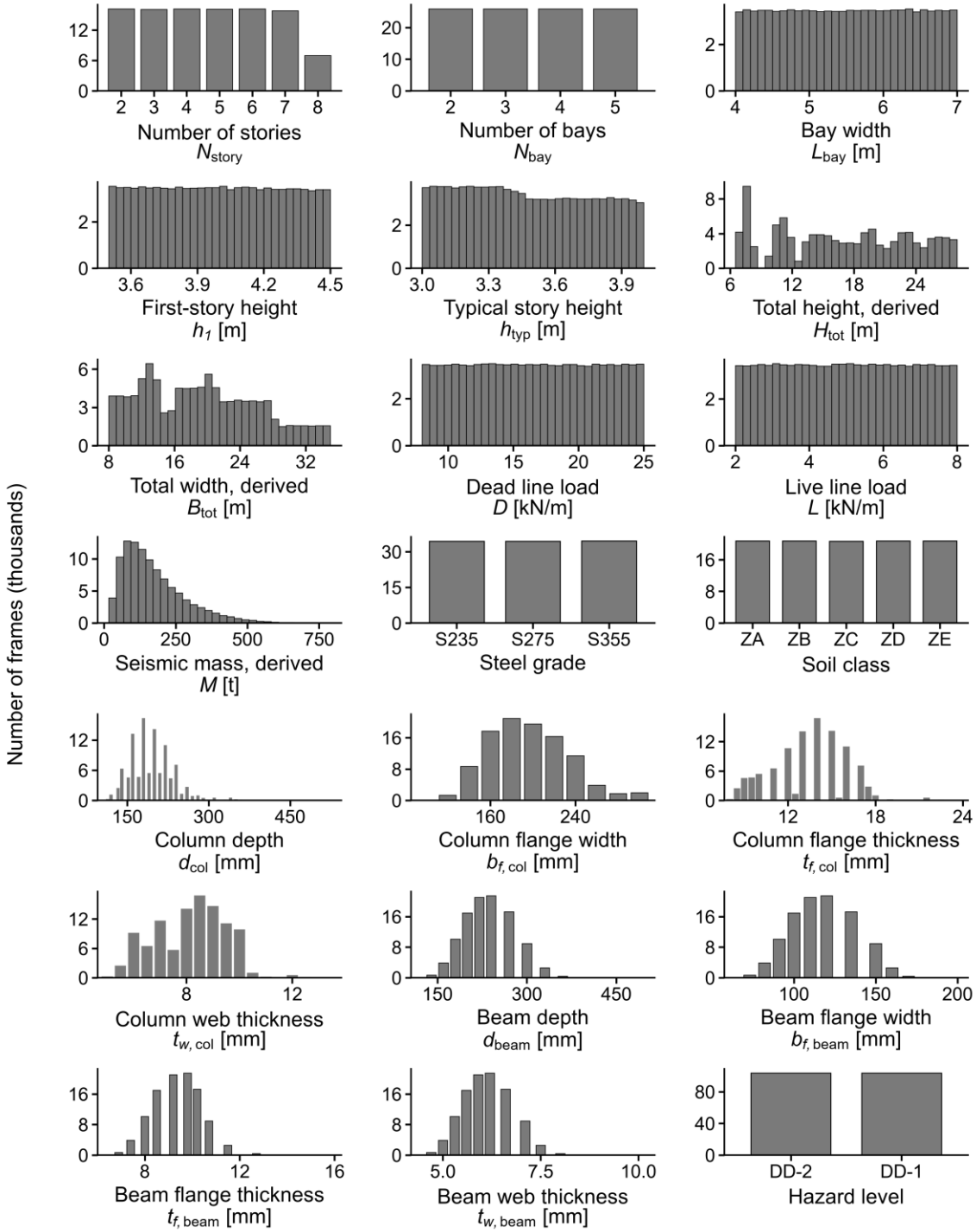


Figure S1. Distributions of the 21 input variables in the database.

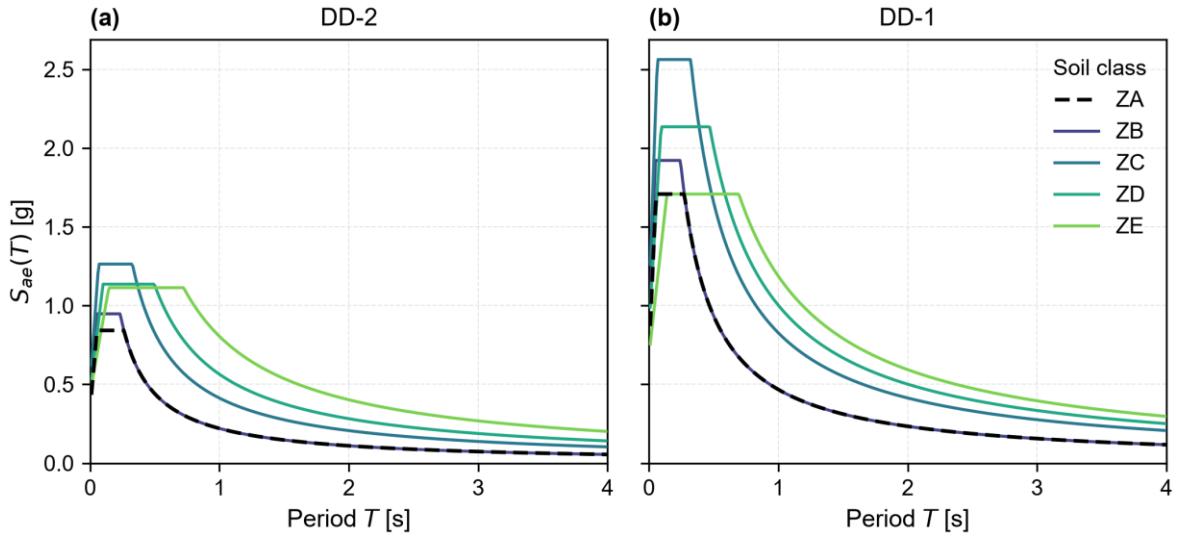


Figure S2. Design spectra of TBEC 2018 at the site: (a) DD-2, (b) DD-1.

S3. Reanalysis of the Member Backbone and of Second-Order Effects

The residual strength ratio κ and the ultimate rotation θ_u are modeling assumptions (Section 2.2), and their effect on the labels was examined on 1,000 randomly drawn frames of the database (seed 2031), that is, 2,000 analyses. These frames were analyzed again with the database generator and their recorded section pairs, first with the production values, which returned all 2,000 labels unchanged, and then once per backbone variant. Five variants were analyzed: κ of 0.30 or 0.50 in place of 0.40, θ_u of 0.12 or 0.24 rad in place of 0.18 rad, and, for every column, the beam-based equations of Section 2.2 replaced by the monotonic parameters proposed by Lignos et al. (2019). For this last variant, the regression equations of that study supply the plastic rotations before and after capping, the capping ratio, the residual ratio and a 0.15 rad ultimate rotation. They take the web slenderness of the section, the weak-axis slenderness of the column and the gravity-state axial-load ratio as inputs, and the column strength stays at $W_p F_{ye}(1 - P/P_{ye})$. For 99.3% of the sampled column members, the slenderness falls within the data ranges of that study. The κ and θ_u values of the variants are checks chosen in this study, not values from a source.

Table S1 counts the labels, formed with the rules of Section 2.3, that differ from those obtained with the production values. Neither κ nor θ_u changes any label, and with θ_u = 0.12 rad one analysis failed to converge and was excluded. With the column parameters of Lignos et al. (2019), 24 labels become less severe, CP to LS in 7 cases and C to CP in 17, and none becomes more severe. For this variant, the database labels are thus on the conservative side.

Table S1. Label changes under backbone variants, 1,000 sampled frames.

Variant	Analyses compared	Labels changed	More severe	Less severe
$\kappa = 0.30$	2,000	0	0	0
$\kappa = 0.50$	2,000	0	0	0
$\theta_u = 0.12$ rad	1,999	0	0	0
$\theta_u = 0.24$ rad	2,000	0	0	0
Column parameters of Lignos et al. (2019)	2,000	24	0	24

A second reanalysis concerned the beyond-peak group of Section 2.3, in which the target displacement lies past the peak of the capacity curve. A random sample of 300 analyses of this group, stratified by hazard level and label with one analysis per frame, was analyzed again with the database generator and without the P-Delta transformation of the columns. In all 300 reanalyses, the peak of the capacity curve moved beyond the target displacement, which shows that the descending branch before the target displacement is produced by second-order effects.

S4. Acceptance Axial Force at the State of Maximum Plastic Rotation

In the database, the acceptance axial force of a column is its force in the gravity state (Section 2.3), whereas TBEC 2018 Annex 5C.0 evaluates it where the plastic rotation is largest. Table S2 compares the labels obtained with the two definitions for all analyses, with the database labels in its rows and the Annex 5C.0 labels in its columns. They disagree for 2,534 of the 207,186 analyses, and in 2,244 of these cases the Annex 5C.0 label is the more severe one. In the test partition, 796 of the 62,152 labels differ. The adopted classifier, scored against the Annex 5C.0 labels, reaches 0.9719 in accuracy, 0.9469 in macro F1 and 0.8428 in CP F1, with an unsafe rate of 1.80%. The corresponding values for the database labels are 0.9799, 0.9624, 0.8903 and 0.914%. Of the 796 test analyses with a changed label, 638 receive the database label from the classifier, and because most changes are toward a more severe state, these predictions raise the unsafe rate against the Annex 5C.0 labels.

Table S2. Database labels against labels with the Annex 5C.0 acceptance axial force.

Database label	IO	LS	CP	C
IO	49,529	109	0	3
LS	9	71,311	785	521
CP	0	139	12,000	826
C	0	3	139	71,812

S5. Tuning, Selection Criteria and Training Decisions

This section collects the tuning settings, selection criteria and training decisions that Sections 2.4 and 2.5 of the article summarize. The formulations of the first stage and of the three classifiers are given in Eqs. (3), (4) and (11)–(18) of the article.

First stage. The elastic net, the Gaussian process regressor and the network were tuned per quantity with 60, 25 and 30 TPE trials, respectively, the network on a subsample of 8,000 analyses. The target transformations T_q of Eq. (3) of the article are $\log(1+V)$ for V_y and V_{max} and $\log(\delta/H_{tot})$ for the displacements, while $V(\delta_t)$ and T_1 are used untransformed. Each regressor received a mean rank over nine criteria, all evaluated on grouped five-fold out-of-fold predictions for the training frames. Three criteria describe the coefficient of determination across the eight quantities, namely its mean, minimum and standard deviation, and three average the errors over the quantities: root-mean-square error, mean absolute error and MdAPE. Two further criteria give the average shares of predictions within 10% and within 20% of the computed value, and the ninth counts predictions that violate the physical order of the quantities. Table 6 of the article shows that the network ranked first on this basis.

Classifiers. Every search ran the TPE (Akiba et al. 2019; Bergstra et al. 2011) and maximized the macro F1 of Eq. (S4) on the frame-grouped inner validation split, which is taken from the training frames alone. Table S3 lists the search spaces, the budgets and the selected values.

Table S3. Hyperparameter search spaces, budgets and selected values.

Model	Hyperparameter	Search space	Trials	Selected value
Multinomial logistic regression	Inverse L2 strength C_{LR}	10^{-3} to 10^{2}, log-uniform	30	11.1
Gaussian process classifier	Kernel family	Squared exponential or Matérn with ν = 0.5, 1.5, 2.5	20	Matérn, ν = 2.5
	Length scale	0.5 to 30, log-uniform		3.74
Multilayer perceptron	Hidden layers	(128), (128, 64), (256, 128), (128, 64, 32)	40	(256, 128)
	L2 penalty	10^{-6} to 10^{-2}, log-uniform		3.31×10^{-3}
	Initial learning rate	10^{-4} to 5×10^{-3}, log-uniform		1.88×10^{-4}
	Activation	Rectified linear or hyperbolic tangent		Rectified linear

Training decisions of the perceptron. Three decisions were taken on the training frames before the test partition was evaluated. Early stopping on a random validation fraction was avoided, since such a fraction would split the two analyses of a frame, and on the inner validation split it also lowered the macro F1 from 0.9605 to 0.9571. Each network was instead trained for at most 400 epochs and stopped earlier when 15 successive epochs improved the

loss by less than 10^{-4}. Averaging the probabilities of five networks with different seeds, Eq. (17) of the article, increased the cross-validated macro F1 on the training frames (grouped, five folds) from 0.9558 to 0.9623 and the CP recall from 0.865 to 0.884, which led to its adoption. Class-balanced sample weights were also tested. In cross-validation, they increased the CP recall from 0.865 to 0.902 but reduced the macro F1 from 0.9558 to 0.9546, so they were not adopted.

S6. Standard Classification Metrics

The article compares the classifiers with standard metrics, which Section 2.6 of the article describes in words and which are defined here. For each state c, the precision Prec_c, the recall Rec_c and the F1 score $F1_c$ are computed on a set of analyses, the test partition unless stated otherwise. They follow from the number of analyses correctly assigned to c, TP_c, the number wrongly assigned to c, FP_c, and the number of reference c analyses assigned to another state, FN_c:

$$\mathrm{Prec}_c = \frac{TP_c}{TP_c + FP_c} \tag{S1}$$

$$\mathrm{Rec}_c = \frac{TP_c}{TP_c + FN_c} \tag{S2}$$

$$F1_c = \frac{2\,\mathrm{Prec}_c \mathrm{Rec}_c}{\mathrm{Prec}_c + \mathrm{Rec}_c} \tag{S3}$$

The accuracy is the share of correctly classified analyses. The macro F1 is the unweighted mean of the four class F1 scores, which gives the rare CP state the same weight as the others:

$$F1_{macro} = \frac{1}{4}\sum_{c=1}^{4} F\,1_c \tag{S4}$$

The four states are ordered and indexed $i = 1$ to 4 from IO to C. The quadratic weighted kappa (QWK) (Cohen 1968) measures the agreement between the predicted and the reference states on this ordinal scale:

$$\kappa_w = 1 - \frac{\sum_{i=1}^{4}\sum_{j=1}^{4} w_{ij} O_{ij}}{\sum_{i=1}^{4}\sum_{j=1}^{4} w_{ij} E_{ij}}, w_{ij} = (i-j)^2 \tag{S5}$$

where O_{ij} is the number of test analyses with reference state i and predicted state j, and E_{ij} is the number expected if the predicted and reference states were independent, obtained from the product of the marginal totals and scaled to the same total. Each disagreement is thus penalized in proportion to the squared distance between the two states.

The probabilities of reaching or exceeding a state, $p_n^{(e)}$ with the outcome $o_n^{(e)}$ of Eq. (21) of the article, are checked with two further metrics. The expected calibration error (ECE) follows the binned definition of Guo et al. (2017). The N test analyses are sorted into 10 equal-width bins of $p_n^{(e)}$, and in each bin b, holding N_b analyses, the mean predicted probability $\overline{p}_b^{(e)}$ is compared with the observed frequency $\overline{o}_b^{(e)}$ of the event:

$$ECE^{(e)} = \sum_{b=1}^{10} \frac{N_b}{N} \left| \overline{o}_b^{(e)} - \overline{p}_b^{(e)} \right| \tag{S6}$$

The Brier score, the mean squared difference between the predicted probability and the outcome, complements the ECE:

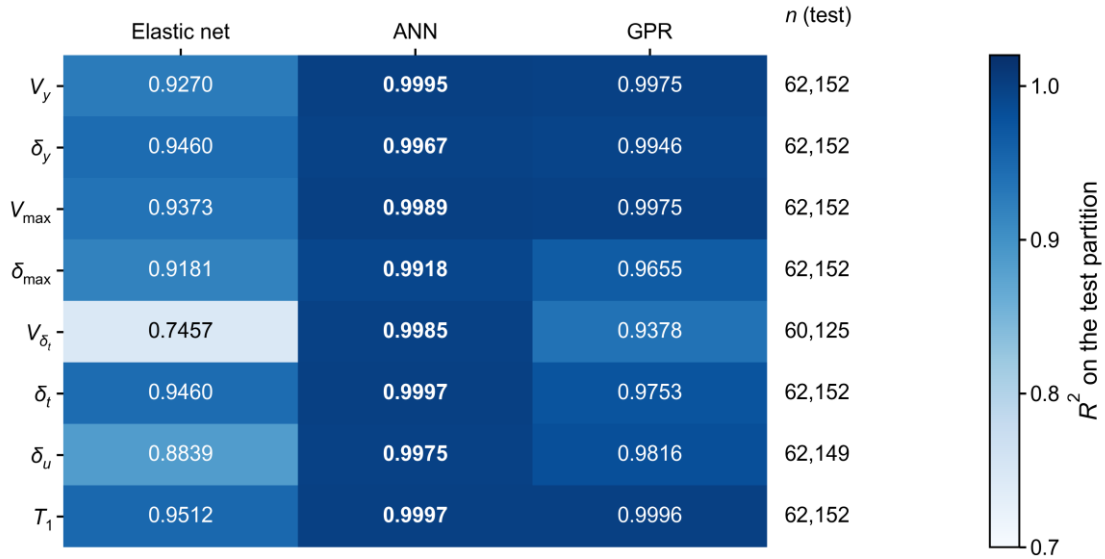
$$BS^{(e)} = \frac{1}{N} \sum_{n=1}^{N} \left(p_n^{(e)} - o_n^{(e)} \right)^2 \tag{S7}$$

S7. First-Stage Accuracy per Quantity

The accuracy of the first stage is resolved here by quantity. Figure S3 gives the test coefficient of determination of the three regressors, the elastic net, the Gaussian process regressor and the network, for each response quantity of Table 3 of the article. The Gaussian process nearly matches the network for the strength quantities and T_1 but trails it for δ_{max}, δ_t, δ_u and $V(\delta_t)$, and the elastic net falls furthest behind for $V(\delta_t)$. Figure S4 plots the predictions of the adopted network against the computed values for the test partition, and Table S4 lists its accuracy. For every quantity, including those on the softening branch, the points stay close to the line of equality over the whole range.

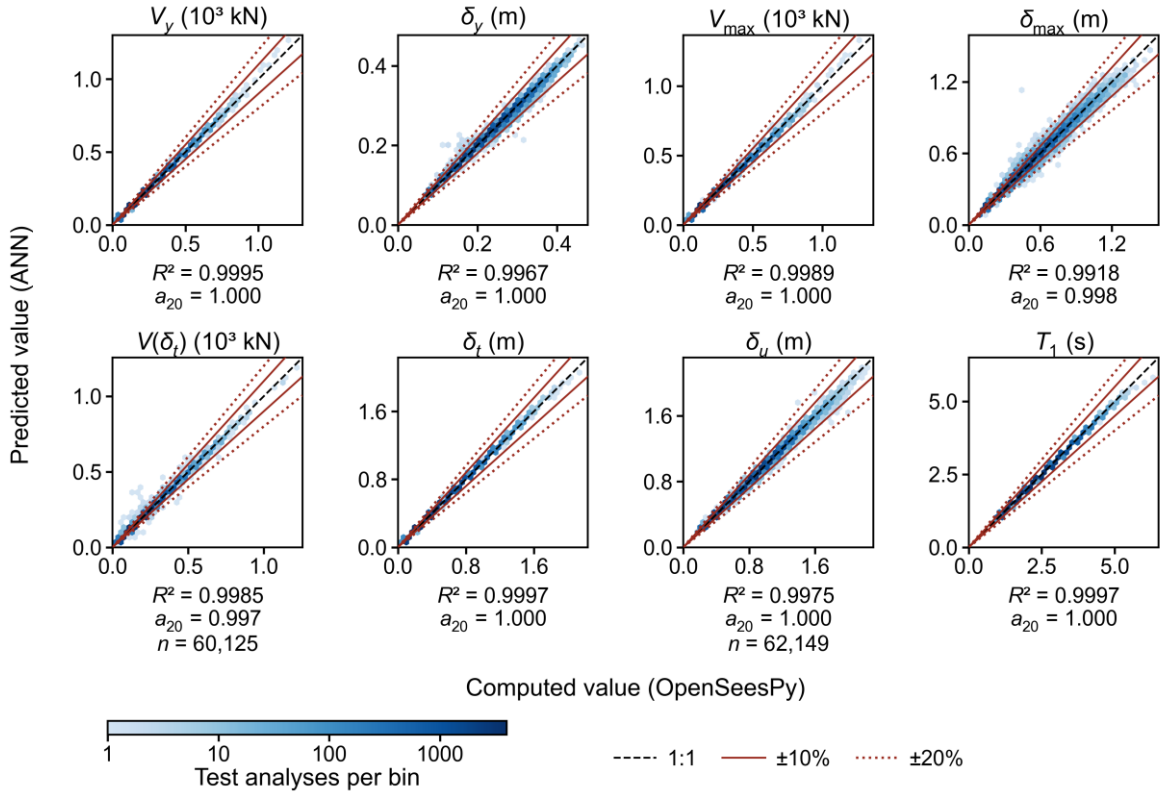
Table S4. Test accuracy of the adopted first-stage network by response quantity.

Quantity	Test analyses	R^2	MdAPE [%]	Share within 20%
V_y	62,152	0.9995	0.70	1.0000
δ_y	62,152	0.9967	0.87	0.9998
V_{max}	62,152	0.9989	1.10	0.9998
δ_{max}	62,152	0.9918	1.78	0.9983
$V(\delta_t)$	60,125	0.9985	0.60	0.9969
δ_t	62,152	0.9997	0.49	1.0000
δ_u	62,149	0.9975	1.16	1.0000
T_1	62,152	0.9997	0.36	1.0000



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155 **Figure S3.** Test R^2 of the three first-stage regressors by response quantity.



156

157 **Figure S4.** Parity plots of the adopted first-stage network for the eight pushover quantities, test
158 partition.

159 ## S8. Paired Differences between the Classifiers

160 The classifier comparison of Section 3.1 is detailed here by paired differences. Table S5 gives
161 the paired bootstrap comparisons of the macro F1. Each difference subtracts the macro F1 of
162 the second classifier from that of the first, both recomputed on the same resampled frames in

each of the 400 resamples. The two-sided p-value doubles the smaller of two shares, that of resamples with a non-positive difference and that of resamples with a non-negative difference, and it cannot fall below 1/400. The Holm procedure then adjusts the three p-values. The smallest attainable value is therefore 0.0025 before and 0.0075 after the adjustment, which all three comparisons reach, so the intervals describe the separation better than the p-values.

Table S5. Paired bootstrap comparison of the macro F1 of the classifiers, test partition.

Comparison	Difference in macro F1	95% interval	p (Holm)
Logistic regression − Gaussian process classifier	0.0243	0.0213 to 0.0271	0.0075
Logistic regression − multilayer perceptron	−0.1726	−0.1765 to −0.1694	0.0075
Gaussian process classifier − multilayer perceptron	−0.1969	−0.2008 to −0.1933	0.0075

S9. Class-Wise Performance

Table S6 lists the class-wise test metrics of the adopted classifier, which Figure 8 of the article shows at the margins of the confusion matrix.

Table S6. Class-wise test performance of the adopted multilayer perceptron.

State	Support	Precision	Recall	F1
IO	14,891	0.9951	0.9874	0.9912
LS	21,787	0.9788	0.9845	0.9816
CP	3,889	0.8908	0.8897	0.8903
C	21,585	0.9867	0.9864	0.9865
Macro average	62,152	0.9629	0.9620	0.9624

S10. Errors by Subgroup

The subgroup error rates of Section 3.4 are listed here with their intervals in Table S7 and plotted in Figure S5. The intervals come from a frame-level bootstrap of the test partition with 2,000 resamples, in which both analyses of a frame stay together. Subgroups are formed by hazard level, story count and soil class, by whether an analysis has a limited-ductility member or a column end above $0.50P_c$ under gravity loads, by membership in the beyond-peak group (Section 2.3) and by the joint-domain distance of Section 2.6. The rates are computed against the reference labels of this study.

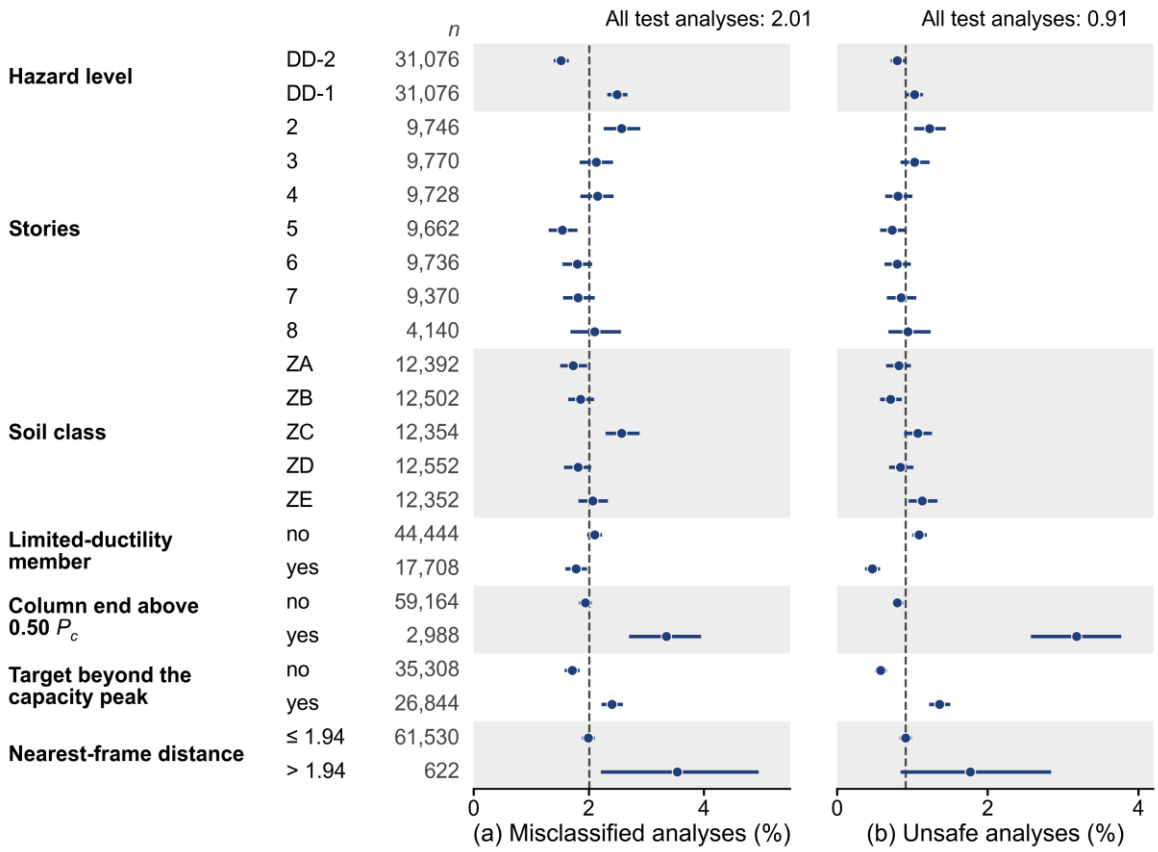


Figure S5. Subgroup error rates of the adopted classifier with 95% intervals, test partition.

Table S7. Error rates of the adopted classifier by subgroup, test partition, with 95% intervals.

Subgroup	Analyses	Misclassified [%]	Unsafe [%]
All test analyses	62,152	2.01 (1.90 to 2.12)	0.91 (0.84 to 1.00)
DD-2	31,076	1.52 (1.39 to 1.65)	0.80 (0.71 to 0.91)
DD-1	31,076	2.49 (2.32 to 2.67)	1.03 (0.91 to 1.14)
2 stories	9,746	2.57 (2.26 to 2.89)	1.23 (1.02 to 1.44)
3 stories	9,770	2.13 (1.84 to 2.42)	1.03 (0.84 to 1.23)
4 stories	9,728	2.15 (1.85 to 2.43)	0.81 (0.64 to 1.00)
5 stories	9,662	1.54 (1.30 to 1.80)	0.73 (0.57 to 0.90)
6 stories	9,736	1.80 (1.54 to 2.06)	0.80 (0.63 to 0.98)
7 stories	9,370	1.81 (1.55 to 2.10)	0.85 (0.66 to 1.05)
8 stories	4,140	2.10 (1.68 to 2.56)	0.94 (0.68 to 1.24)
Soil class ZA	12,392	1.73 (1.50 to 1.97)	0.82 (0.65 to 0.98)
Soil class ZB	12,502	1.86 (1.64 to 2.09)	0.71 (0.57 to 0.86)
Soil class ZC	12,354	2.57 (2.29 to 2.88)	1.07 (0.89 to 1.26)
Soil class ZD	12,552	1.81 (1.57 to 2.04)	0.84 (0.69 to 1.01)
Soil class ZE	12,352	2.07 (1.82 to 2.33)	1.13 (0.95 to 1.33)
Limited-ductility member: no	44,444	2.10 (1.97 to 2.23)	1.09 (1.00 to 1.19)
Limited-ductility member: yes	17,708	1.78 (1.59 to 1.97)	0.47 (0.37 to 0.57)
Column end above 0.50P_c: no	59,164	1.94 (1.83 to 2.05)	0.80 (0.73 to 0.88)
Column end above 0.50P_c: yes	2,988	3.35 (2.70 to 3.95)	3.18 (2.57 to 3.77)
Beyond-peak group: no	35,308	1.71 (1.58 to 1.84)	0.58 (0.50 to 0.66)
Beyond-peak group: yes	26,844	2.40 (2.22 to 2.59)	1.36 (1.22 to 1.50)
Joint-domain distance up to 1.94	61,530	1.99 (1.88 to 2.10)	0.91 (0.83 to 0.99)
Joint-domain distance above 1.94	622	3.54 (2.21 to 4.95)	1.77 (0.84 to 2.84)

S11. Group-Held-Out Extrapolation Tests

The group-held-out runs of Section 3.4 are listed here together with the accuracy of the refitted first stage, and Table S8 gives the values that Figure 12(a) of the article plots, with the CP recall in addition. In each run, every analysis of the named group was removed from the training partition, both stages were refitted without the group, and the analyses of the group in the test partition of the main split were evaluated. No physical frame is therefore shared between training and evaluation. When a hazard level is held out, both stages were trained on the analyses of the training frames at the remaining level. The last column reports the MdAPE of the refitted first stage for δ_t in the evaluated analyses.

Table S8. Group-held-out extrapolation tests.

Held-out group	Test analyses	Macro F1	CP recall	Unsafe [%]	First-stage MdAPE of δ_t [%]
None (main split)	62,152	0.9624	0.8897	0.914	0.49
2 stories	9,746	0.8079	0.3971	2.247	1.60
3 stories	9,770	0.9350	0.8321	2.385	0.86
4 stories	9,728	0.9546	0.8877	1.439	0.42
5 stories	9,662	0.9625	0.8951	1.076	0.45
6 stories	9,736	0.9201	0.7123	0.236	0.63
7 stories	9,370	0.9506	0.8728	1.291	0.60
8 stories	4,140	0.9290	0.7346	0.870	0.51
Soil classes ZA and ZB	24,894	0.5696	0.0684	3.772	43.11
Soil class ZC	12,354	0.5972	0.1883	29.488	18.51
Soil classes ZD and ZE	24,904	0.5434	0.0355	1.594	26.44
Hazard level DD-2	31,076	0.2812	0.0000	0.058	102.40
Hazard level DD-1	31,076	0.1948	0.0025	73.674	50.96

S12. Global SHAP Ranking and Period Elongation

Figure S6 shows the global attribution ranking of Section 3.5 for all 26 features: the mean absolute SHAP values of each feature are summed over the four states, and each bar shows how its total divides among the states. The kernel SHAP explanations of Section 2.6 used a k-means summary of 50 training analyses as background and 1,000 model evaluations per explained analysis. The directions cited in Section 3.5 follow the sign of the rank correlation between the value of a feature and its attribution to each state over the explained analyses. The period elongation in Section 3.5 is the median, by story count, of the ratio between the first-mode period in the gravity state and the period from a modal analysis before the gravity load is applied, 1.044 for two stories and 1.084 for eight stories.

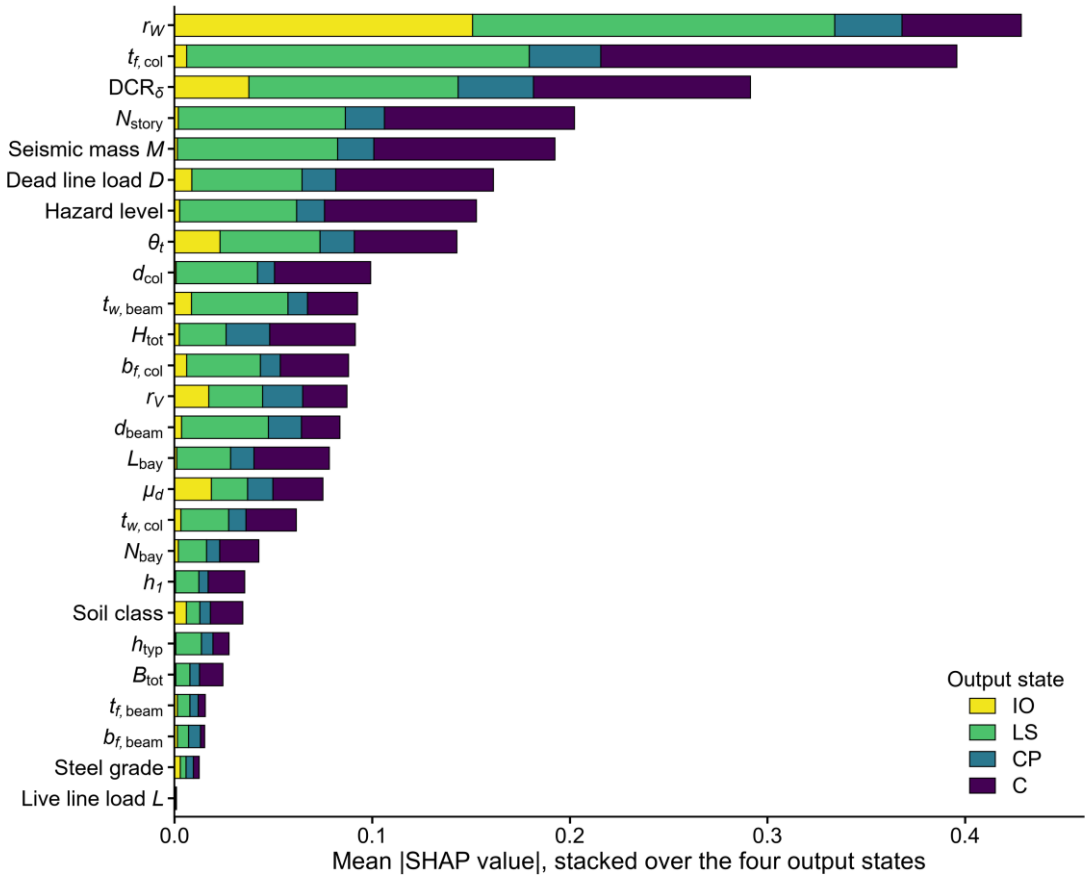


Figure S6. Mean absolute SHAP values of all 26 features, stacked by performance state.

S13. Desktop Application

The trained framework is packaged as a desktop application for screening within the domain of the database. Figure S7 shows this application, named Performance State Predictor for Steel Moment Frames, which runs the adopted two-stage model. As inputs, the user enters the story and bay counts, the bay width, the first-story and typical story heights, the dead and live line loads D and L, the soil class, the steel grade and the column and beam profiles. The derived inputs, namely total height, total width, seismic mass and the eight section dimensions, are then computed with the formulas of the database generator. The hazard level is not an input, as both levels are always evaluated. For each level, the output contains the predicted state, the probabilities of all four states, the eight first-stage quantities and the five ratios, together with a plot that joins the predicted characteristic points by straight segments and is marked as not being an analysis result.

Frame

N_{story}	5
N_{bay}	3
L_{bay} [m]	5.030
h_1 [m]	4.033
h_{typ} [m]	3.210
D [kN/m]	21.63
L [kN/m]	6.90
Steel grade	S275
Soil class	ZC
Column	HE220B
Beam	IPE240

Derived inputs

H_{tot} [m]	16.87	B_{tot} [m]	15.09
M [t]	188.8	W [kN]	1,788

d / b / t_f / t_w [mm]

Column	HE220B: 220 / 220 / 16 / 9.5
Beam	IPE240: 240 / 120 / 9.8 / 6.2

Validity checks

All inputs inside the Table 2 ranges

Ductility, column	**high**
Ductility, beam	**high**
Axial load P_u/P_{ye}	**pass**
Gravity P_g/P_c, ground	**pass**
Beam gravity flexure	**pass**
Clear lengths	**pass**
Total height	**pass**
Nearest database frame	**1.12 ≤ 1.94**

Not checked (need an analysis):

first-mode mass ratio	not checked
column P/P_c at acceptance	not checked

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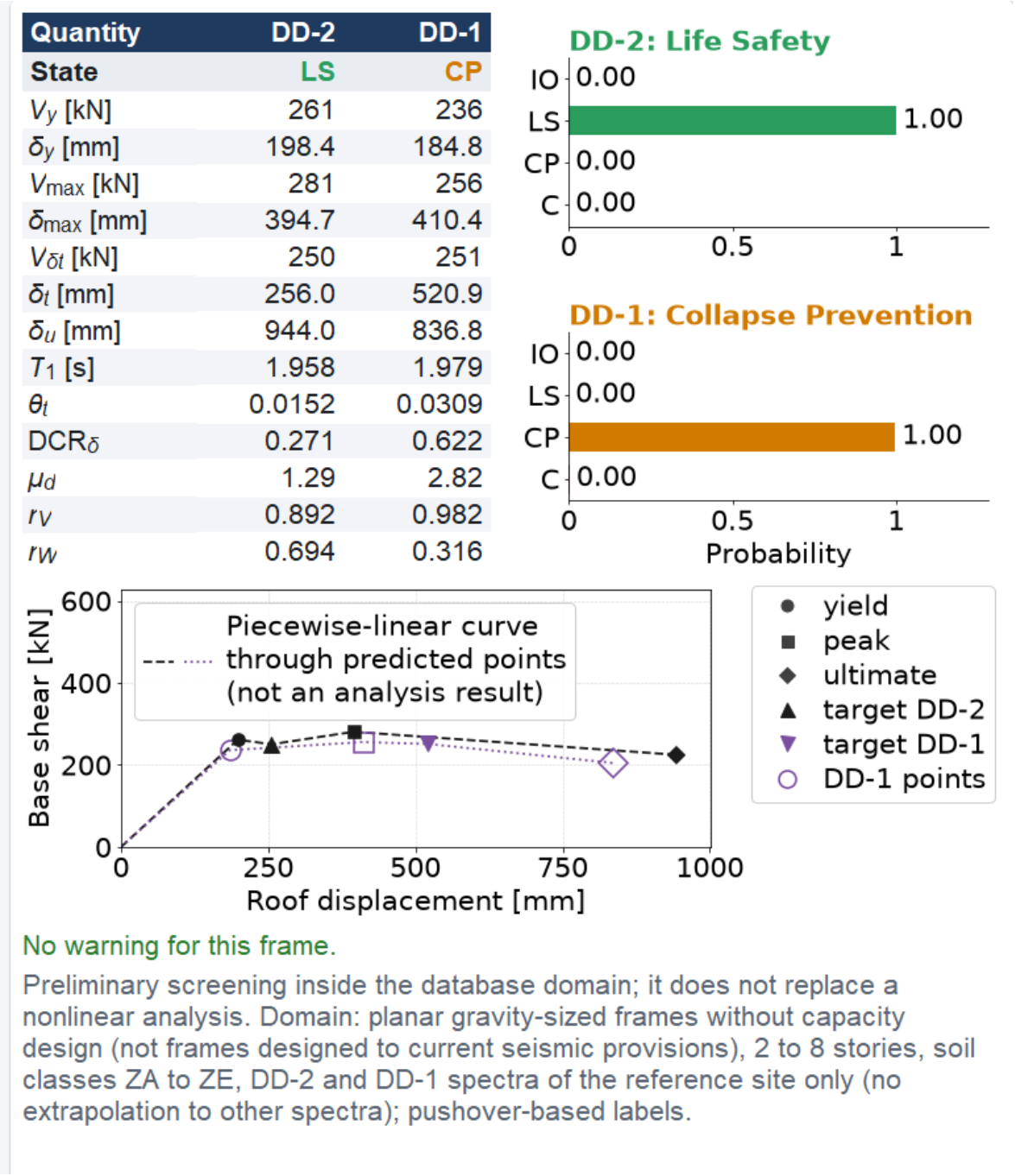


Figure S7. Desktop application evaluating a test frame of the database: (a) input panel, (b) output panel.

Three checks accompany each prediction. A validity panel compares every input with its range in Table 2 and repeats, for the chosen profiles, the closed-form screens on the sections and the 28 m height limit of Section 2.1. The application cannot verify the first-mode effective mass ratio or the axial-load ratio of the columns under the acceptance axial force without an analysis, and it reports both as unchecked. A joint-domain check standardizes the 20 frame inputs with the training statistics and measures the distance from the frame to the nearest training frame. A distance above 1.94, the 99th percentile of the test-frame distances,

triggers a warning. A frame can pass every range check and still be flagged, because the database sections depend on the frame variables, and flagged test frames are misclassified more often (Section 3.4). Further warnings appear when the predicted yield base shear exceeds the predicted peak base shear, when the target displacement exceeds the predicted ultimate displacement, or when predicted points are out of physical order.

The trained networks are evaluated without the training library. The reported first-stage values, ratios and class probabilities of all 62,152 test analyses are reproduced with relative differences under 10^{-12}, and the reported states exactly. The application is restricted to preliminary screening of frames inside the database domain, namely planar gravity-sized frames without capacity design, with two to eight stories, on soil classes ZA to ZE and under the two spectra of the site (Section 3.4). It does not take the place of a nonlinear analysis of an individual building.

S14. Computational Demand

The computational effort is concentrated in two one-time tasks: the generation of the database and the training of the models. For each of the 207,186 analyses, the generator assembles the model, runs the gravity analysis, solves the eigenvalue problem in the gravity state and carries the pushover beyond the peak. On a computer with a 12-core AMD Ryzen 9 7845HX processor, the 113,147 draws of the production run took 501.7 min with 20 worker processes, and the median wall-clock time per frame in a worker process was 3.2 s (mean 5.4 s).

Training the classifiers costs much less. Leaving aside the hyperparameter search and the first stage, fitting took 2,648.9 s for the five networks of the adopted perceptron together, 7.2 s for the Gaussian process classifier, which uses a balanced subsample, and 3.8 s for logistic regression. With all tuning and evaluation experiments, the complete machine learning run lasted 13.3 h on 12 worker processes of the same computer.

After training, the two-stage model in the implementation of the desktop application processed the 62,152 test analyses in 7.5 s of wall-clock time, or 0.12 ms per analysis, and a single frame at both hazard levels in 1.7 ms. This is about three orders of magnitude faster than the median analysis of a frame, without counting the one-time cost of database generation and training.

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